

DAM Governance Intelligence Platform

AI-Assisted Governance Workflow & Document Orchestration for Danantara Asset Management

CLIENT

BPI Danantara — Danantara Asset Management (DAM)

DELIVERY PARTNER

IBM

DOCUMENT STATUS

Draft v1.0

DOCUMENT DATE

June 2026

SCOPE

Short-Term Quick Wins + Long-Term Platform Vision

CLASSIFICATION

Confidential — Internal Use Only

TABLE OF CONTENTS

1. Executive Summary
2. Problem Statement & Context
3. Goals & Success Metrics
4. Stakeholders & User Personas
5. Pain Point Analysis
6. Product Scope
7. Functional Requirements — Phase 1 (Quick Wins)
8. Functional Requirements — Phase 2 (Platform Expansion)
9. Non-Functional Requirements
10. System Architecture Principles
11. Data & Security Governance
12. Risk Register
13. Out of Scope
14. Open Questions & Assumptions
15. Appendix — Feature Inventory & Prioritization

1 Executive Summary

Danantara Asset Management (DAM) is Indonesia's sovereign wealth fund governance gatekeeper, mandated to review, validate, and recommend corporate actions submitted by state-owned enterprise (SOE) subsidiaries. This function requires the production of multiple, interdependent governance artefacts — including NOTA documents, board notes, decision papers, and formal communications — all under strict Standard Operating Procedure (SOP) compliance.

The current operating model is predominantly manual. Senior analysts spend an estimated 30% of their time on cross-document consistency checks alone. Drafting is repetitive, fragmented across teams, and exposed to significant version-control risk. The absence of a controlled AI environment has led teams to adopt public AI tools informally, creating data confidentiality and auditability gaps.

Target outcome (Phase 1): Achieve a minimum 30% productivity improvement in governance document preparation through AI-assisted drafting, cross-document consistency validation, and structured workflow automation — while maintaining full SOP compliance and personal liability protection.

IBM has been engaged to design and deliver the DAM Governance Intelligence Platform: a secure, enterprise-controlled system that digitizes SOPs, automates low-risk document workflows, assists human judgment on complex matters, and establishes a single source of truth across the governance artefact lifecycle.

This document defines the product requirements for both the short-term quick-win phase and the long-term BPI-wide governance platform vision.

2 Problem Statement & Context

2.1 Organizational Context

DAM operates as the central governance gatekeeper within the BPI Danantara ecosystem. Corporate action requests from SOE subsidiaries flow through DAM for review, evaluation, and approval recommendation before escalation to boards or regulators. Each request triggers a cascade of governance documents that must be drafted, reviewed, approved, and filed in strict sequence.

2.2 The Governance Document Lifecycle

Each corporate action review currently requires the parallel production of:

- **NOTA documents** — detailed internal review notes (80% descriptive/restatement, 20% analytical judgment)
- **Board notes** — concise summaries prepared for board-level review
- **Decision documents** — formal records of the governance outcome
- **Communications** — official responses dispatched to the requesting SOE subsidiary

These artefacts must be internally consistent and SOP-compliant. Currently, they are drafted independently by different team members, creating structural risk of divergence.

2.3 Current State Quantification

Metric	Current State Observation
Senior analyst time on consistency checking	~30% of total working hours
NOTA content that is descriptive / restatement	~80% (copy, summarize, restate from submissions)
NOTA content requiring judgment	~20% (analysis, risk assessment, recommendation)
AI tool usage	Public AI tools in active informal use (uncontrolled)
Document storage	OneDrive with unclear data governance policy
SOP completeness	Multiple SOPs; some incomplete or still evolving
Unnecessary approvals processed	Present — routing is partly judgment-based, not rule-based

3 Goals & Success Metrics

3.1 Business Goals

- Reduce redundant manual drafting effort by automating descriptive/restatement content in governance documents
- Eliminate cross-document inconsistency as a source of governance risk and senior review burden
- Enforce SOP routing rules programmatically to prevent unnecessary or incorrect approval pathways
- Establish a controlled, enterprise-grade AI environment replacing informal public AI tool use
- Reduce personal liability exposure for individual officers by codifying rules into the system and removing humans from low-risk process steps

3.2 Key Success Metrics

Metric	Baseline	Phase 1 Target	Notes
Overall governance document preparation time	Current baseline	-30% minimum	Mandated quick-win target
Senior analyst time on consistency checking	~30%	<10%	Cross-document checker feature
NOTA first-draft generation time	Manual hours	AI draft in <5 minutes	Descriptive sections only
Cross-document inconsistency rate at sign-off	Reported as significant	Near zero (system-flagged)	Automated before human review
Unnecessary approval routing incidents	Present	Eliminated for rule-based cases	SOP routing engine
Controlled AI environment adoption	0% (public tools in use)	100% team migration	Sandbox deployment
Audit trail completeness	Absent	100% of actions logged	Non-functional requirement

4 Stakeholders & User Personas

4.1 Primary Stakeholders

Stakeholder	Role	Primary Interest
DAM Senior Leadership	Approving officers and governance decision-makers	Reduced time on consistency checks; confidence in document integrity; reduced personal liability
DAM Governance Analysts	Primary drafters of governance artefacts	AI-assisted first drafts; clear SOP routing guidance; version control
DAM Legal & Compliance	SOP owners; final sign-off on document integrity	SOP compliance enforcement; audit trail; liability documentation
IBM Delivery Team	Solution architect and delivery partner	Clear, stable requirements; defined integration boundaries
SOE Subsidiaries	Submission originators; recipients of formal decisions	Faster turnaround; transparent status tracking
BPI Danantara IT / Security	Infrastructure and data governance owners	Data residency; no external training; auditability; access control

4.2 Key User Personas

Persona A — The Governance Analyst

- **Profile:** Mid-level professional responsible for ingesting SOE submissions, drafting NOTA documents, and coordinating the artefact lifecycle
- **Primary pain:** Spends majority of time on copy-summarize-restate tasks; must manually cross-check multiple documents before senior review
- **Key needs:** AI-generated first draft for descriptive sections; template-guided structure; automatic inconsistency flagging between artefacts
- **Fear:** Producing a document with an error that creates personal liability

Persona B — The Senior Officer / Approver

- **Profile:** Director-level officer who reviews governance artefacts and makes final recommendations to boards
- **Primary pain:** 30% of time consumed by checking that documents are consistent with each other; manual reconciliation before sign-off
- **Key needs:** Confidence that descriptive content is pre-validated; ability to focus exclusively on judgment sections; clear audit trail for accountability
- **Fear:** Personal liability from signing off on a flawed governance artefact

Persona C — The SOP / Compliance Officer

- **Profile:** Legal or compliance professional who owns SOPs and is accountable for institutional process integrity
- **Primary pain:** SOPs exist in multiple forms, some incomplete; routing decisions rely partly on individual judgment, creating inconsistency
- **Key needs:** Digitized SOPs enforced by the system; decision routing engine; SOP version control with audit log
- **Fear:** A governance decision made on the wrong SOP pathway creating regulatory exposure

5 Pain Point Analysis

Pain Point A — Inconsistency and Multiple Versions of Truth

Impact: Senior officers spend approximately 30% of their time on manual reconciliation and consistency checking across parallel governance artefacts.

Multiple team members draft NOTA documents, board notes, decision papers, and communications in parallel, referencing the same submission package independently. Source document inconsistencies (e.g. a figure or date that differs between the original submission and an internal attachment) propagate through all downstream artefacts. Final sign-off becomes a manual reconciliation exercise rather than a judgment exercise.

Pain Point B — Over-Documentation Driven by Liability Avoidance

SOP compliance is non-negotiable due to legal exposure risk at both institutional and individual level. This compliance imperative, while necessary, has produced a culture of defensive documentation where artefacts are maximally detailed regardless of case complexity. The result is redundant duplication of content and governance documents that function primarily as liability shields rather than decision-support tools.

Pain Point C — Fragmented Workflow and Unclear SOP Routing

DAM operates under multiple SOPs, some of which are incomplete or still evolving. The selection of the applicable SOP pathway for a given corporate action request is sometimes clear and rule-based, but often relies on individual analyst judgment. This leads to two failure modes:

- **Under-routing:** necessary approvals missed because the analyst judged a pathway incorrectly
- **Over-routing:** approvals sought that are not required, adding processing delays without governance value

Pain Point D — Manual, Repetitive Drafting Burden

Approximately 80% of NOTA document content is descriptive: copying, summarizing, and restating information already present in the SOE submission package. Only the remaining 20% requires analytical judgment unique to the reviewer. This ratio means the majority of analyst drafting time is spent on mechanical, high-volume, low-value tasks that are highly amenable to automation.

Pain Point E — Absence of a Controlled AI Environment

Governance analysts are already using public AI tools informally to address the drafting burden described above. This creates three institutional risks:

- **Hallucination risk:** public AI tools may generate plausible but factually incorrect content; in a governance context this is a critical failure mode
- **Confidentiality leakage:** sensitive SOE submission data may be transmitted to third-party AI providers
- **Auditability gap:** no record of what AI-assisted content was included in signed governance artefacts

Pain Point F — Personal Liability Exposure (Critical Differentiator)

Unlike many institutional governance frameworks, DAM officers bear individual personal liability — not merely institutional accountability — for governance decisions. This creates a perverse dynamic: greater involvement in a governance process increases an individual's liability exposure. The desired outcome is to codify low-risk process rules into the system so that human involvement is required only where genuine judgment is exercised, thereby protecting individuals from liability on process-mechanical matters.

6 Product Scope

6.1 In Scope

- AI-assisted governance document drafting engine (descriptive sections of NOTA and related artefacts)
- Cross-document consistency checker across the governance artefact set
- Version control and master document hierarchy with propagation logic
- Rule-based SOP eligibility and routing engine
- Secure enterprise AI sandbox (no external data transmission; no external model training)
- Integrated workflow and document orchestration engine linking the full submission-to-communication lifecycle
- SOP digitization and decision-tree encoding
- Risk-based automation segmentation (fully automated low-risk flows; human-in-the-loop for complex matters)
- Long-term: BPI-wide enterprise governance platform across DAM, DIM, and DSI entities
- Long-term: Agentic governance layer with draft, validate, route, and flag capabilities

6.2 Phasing

Phase	Label	Focus
Phase 1	Quick Wins	8 capability modules targeting 30% productivity improvement within DAM
Phase 2	Platform Expansion	BPI-wide governance platform and agentic governance layer across DAM, DIM, DSI

7 Functional Requirements — Phase 1 (Quick Wins)

FR-1 — AI-Assisted Drafting Engine

P1 — CRITICAL

User story: As a governance analyst, I want the system to generate a first draft of the descriptive sections of a NOTA document based on the SOE submission package and historical precedents, so that I can focus my time on the analytical judgment sections.

Functional Requirements

- The system shall ingest structured SOE submission documents (PDF, DOCX, and any DAM-standard format) as input context for draft generation.
- The system shall generate first-draft text for all descriptive sections of the NOTA template (background, request summary, entity information, submission restatement) with full traceability to source paragraphs.
- The system shall not autonomously generate content for judgment sections (risk assessment, recommendation, conditions) — these shall remain blank with a human-assignment prompt.
- The system shall surface the source excerpt alongside each generated paragraph so the analyst can verify accuracy before acceptance.
- The system shall support analyst acceptance, rejection, or manual editing of each AI-generated paragraph as discrete tracked actions.
- All accepted, rejected, and edited AI-generated content shall be logged with a timestamp and user identifier for auditability.
- The system shall use historical approved NOTA documents as fine-tuning or retrieval-augmented generation (RAG) context to improve drafting alignment with DAM style and precedent.

Performance & Quality Metrics

The following metrics govern ongoing measurement of drafting engine quality and shall be tracked per model version and reported on the IBM delivery scorecard at each sprint review.

Metric	Definition	Target Threshold	Measurement Method	Notes
--------	------------	------------------	--------------------	-------

Draft Generation Latency	Wall-clock time from submission package upload to full first draft available in the analyst's editing view	<5 min (standard); <10 min (large package)	System timestamp; P95 latency measured across all draft requests in a 30-day window	Standard = submission package ≤50 pages. Large = 51–150 pages. Packages above 150 pages require a separate SLA negotiation.
Descriptive Section Coverage Rate	Percentage of NOTA descriptive sections for which the engine produces at least one draft paragraph	≥95%	Automated comparison of generated output against the NOTA template section list per case; logged at draft creation	A section is "covered" if one or more draft paragraphs are generated. Gaps may occur if the submission package omits required source data — these shall be flagged as "Insufficient source material" rather than engine errors.
Source Traceability Rate	Percentage of AI-generated sentences that carry a citation link to a retrievable passage in the source submission package	100%	Automated validation at generation time; any sentence lacking a source link is blocked from rendering in the analyst's view	Non-negotiable hard requirement. The system shall never render an unsourced generated sentence to the analyst. If a sentence cannot be grounded in a source passage, it is suppressed and a "Low confidence — manual input required" placeholder is shown instead.

Analyst Acceptance Rate	Percentage of AI-generated paragraphs accepted by the analyst without any edit	≥60% at pilot launch; ≥75% at 3 months post-launch	Ratio of "Accept" actions to total paragraphs generated, aggregated per analyst per month; reported on the management dashboard	This is a quality proxy, not a hard system gate. A low acceptance rate signals model drift or submission format changes and triggers a model refresh review. Individual analyst acceptance rates below 40% shall trigger a targeted training session.
Post-Edit Distance (Analyst Edit Burden)	Average number of word-level edits applied by the analyst to accepted AI paragraphs before final submission	<15% word change rate on accepted paragraphs at 3 months post-launch	Diff computed between AI-generated paragraph and analyst-final paragraph at submission time; word change rate = $\frac{\text{changed words}}{\text{total words}}$	Tracks how much the analyst must correct accepted content. A rising edit distance signals model quality regression and triggers a root-cause review. Measured monthly; reported per section type (background, entity info, etc.).
Judgment Section Contamination Rate	Number of instances per month in which AI-generated content appears in a judgment section without explicit analyst authorship	Zero (hard block)	Automated scan of all submitted NOTAs: any judgment section containing text not logged as analyst-authored triggers a system alert and incident record	Critical compliance metric. Any non-zero occurrence in a production environment is classified as a Severity 1 incident and triggers an immediate system review by the IBM team and DAM compliance officer.

Audit Log Completeness	Percentage of draft generation events for which a complete audit log record exists (user ID, timestamp, model version, paragraphs generated, paragraphs accepted/rejected/edited)	100%	Monthly reconciliation: count of submitted NOTAs with AI involvement vs. count of corresponding complete audit records; any gap is flagged as a data integrity incident	Partial audit records (missing any required field) are treated as absent for compliance purposes.
RAG Retrieval Relevance Score	Average semantic similarity score between the retrieved source passage and the generated paragraph it grounds	≥ 0.75 (cosine similarity on the embedding model used)	Computed at generation time for every paragraph; logged alongside the draft; P10 score (worst-decile) must not fall below 0.60	A falling relevance score indicates that the retrieval model is surfacing weakly related passages, which increases hallucination risk. Triggers a retrieval index refresh when the 30-day P10 drops below 0.60.

Measurement Methodology

- **Baseline period:** Metrics shall be collected from the first day of pilot deployment. A formal baseline snapshot shall be taken at the end of week 2 of the pilot before any model adjustments are made.
- **Reporting cadence:** Metrics dashboard updated daily; formal IBM delivery scorecard review monthly. Any metric breaching its threshold triggers an out-of-cycle review within 5 business days.
- **Test set:** A curated set of 20 historical NOTA cases (with known correct outputs, anonymized) shall be maintained as a regression test suite. The drafting engine is run against this set before every model update; performance must not regress more than 3 percentage points on any metric before the update is approved for production.
- **Analyst survey:** A structured 5-question satisfaction survey shall be administered to all pilot analysts at week 4 and month 3. Results feed into the qualitative component of the 30% productivity target assessment.

Acceptance Criteria

The following criteria must all be satisfied before FR-1 is marked as accepted at the Phase 1 go-live gate:

#	Acceptance Criterion	Test Scenario	Pass Condition	Verification Owner
A C -1 .1	Draft generation latency is within SLA for a standard-size submission	Upload a complete 40-page SOE submission package and trigger draft generation; record wall-clock time to full draft rendered in analyst view	Draft available in <5 minutes across 5 consecutive test runs	IBM QA / DAM Analyst
A C -1 .2	All descriptive sections are covered by the draft	Run the engine against 10 test submission packages spanning at least 3 NOTA template types; verify section coverage for each	≥95% of template sections covered across the test set; any gap is accompanied by an "Insufficient source material" flag	IBM QA
A C -1 .3	Every generated sentence carries a traceable source link	Inspect 3 generated drafts; for each sentence, follow the source citation link and verify it resolves to the correct passage in the submission package	100% of sentences carry a working source link; zero unsourced sentences rendered to the analyst view	DAM Compliance Officer

A C -1 .4	Judgment sections remain blank and prompt human authorship	Submit 5 test packages; inspect the generated draft and verify that risk assessment, recommendation, and conditions sections contain no AI-generated text	Zero AI-generated content in any judgment section across all 5 test cases; human-authorship prompt is present and visible	DAM Compliance Officer / IBM QA
A C -1 .5	Analyst accept / edit / reject actions are all captured in the audit log	Perform a scripted sequence of accept, edit, and reject actions on a test draft; retrieve the corresponding audit log and verify completeness	Audit log contains one record per action with correct user ID, timestamp, paragraph ID, action type, and final content; no records missing	IBM QA / DAM IT Audit
A C -1 .6	RAG retrieval relevance score meets the minimum threshold	Run the engine against the 20-case regression test set; extract retrieval relevance scores for all generated paragraphs	Average cosine similarity ≥ 0.75 ; P10 ≥ 0.60 across the test set	IBM AI Engineering

A	Pilot analyst satisfaction score meets	Administer week-4 pilot	Average	IBM Delivery
C	threshold	satisfaction survey to	rating ≥ 3.5	Lead / DAM
-1		all pilot analysts	out of 5 on	HR
.7		(minimum 5	overall	
		respondents)	usefulness;	
			$\geq 80\%$ of	
			respondents	
			report the tool	
			saves them	
			meaningful	
			time on	
			descriptive	
			drafting	

FR-2 — Cross-Document Consistency Checker

P1 — CRITICAL

User story: As a senior officer preparing to sign off on a governance file, I want the system to automatically compare all artefacts in the file and flag any inconsistencies, so that I am reviewing judgment rather than performing manual reconciliation.

Functional Requirements

- The system shall compare the following artefact set against a designated master document: NOTA, board note, decision document, and formal communication letter.
- Comparison shall identify: factual inconsistencies (numeric values, dates, entity names, amounts), outdated references (references to a prior document version), and missing updates (a change in the NOTA not reflected in the board note).
- The system shall produce a consistency report listing each flagged discrepancy with: (a) the source artefact and location, (b) the target artefact and location, (c) the type of inconsistency, (d) the severity level, and (e) the recommended resolution action.
- The system shall assign a severity level (Critical / Warning / Informational) to each flagged item according to the severity classification rules defined below.
- The consistency check shall be triggerable on-demand by any team member and shall run automatically when a file is submitted to the senior review queue — the file shall not reach the senior review inbox until the check completes and the report is acknowledged by the submitting analyst.
- The system shall log each consistency check run with the triggering user, timestamp, version of each artefact checked, check duration, number of items flagged by severity, and analyst acknowledgement action.
- Analysts and senior officers shall be able to mark individual flagged items as "Resolved", "Accepted as-is with justification", or "Deferred", and each such action shall be logged with the user identity,

timestamp, and any free-text justification provided.

- The system shall block file progression to the board preparation stage if any unresolved Critical-severity item remains in the consistency report.

Inconsistency Type Taxonomy

The checker shall detect and classify discrepancies across the following defined types. Each type has a default severity that may be overridden by DAM compliance officers via configuration.

Inconsistency Type	Definition	Default Severity	Example
Numeric Value Mismatch	A financial figure, percentage, count, or threshold appears with a different value in two or more artefacts	Critical	NOTA states investment value as IDR 500 billion; board note states IDR 450 billion
Date Inconsistency	A referenced date (submission date, effective date, approval deadline, reporting period) differs across artefacts	Critical	NOTA references submission date as 3 June 2026; decision document references 5 June 2026
Entity Name Mismatch	The legal name, abbreviation, or identifier of an SOE subsidiary, counterparty, or regulatory body differs across artefacts	Critical	NOTA uses "PT Pertamina Hulu Energi"; board note uses "PT PHE"
Stale Version Reference	An artefact references a document version that has since been superseded (e.g. references "NOTA v1" when NOTA v2 is the current master)	Warning	Board note cites "as per NOTA dated 28 May 2026" after NOTA was revised on 3 June 2026
Missing Propagated Update	A field or statement was updated in the master document (NOTA) but the corresponding section in a dependent artefact was not updated to reflect the change	Warning	NOTA risk rating revised from "Moderate" to "High" after amendment; board note still reads "Moderate"

Contradictory Recommendation	The stated governance recommendation or approval condition differs between the NOTA and the decision document	Critical	NOTA recommends "conditional approval subject to 3 conditions"; decision document states "unconditional approval"
Scope Boundary Deviation	The described scope, geography, asset class, or transaction perimeter differs between artefacts	Warning	NOTA scope covers "domestic assets only"; communication letter references "domestic and regional assets"
Structural Omission	A required section present in the NOTA (e.g. risk summary, SOP reference, delegation authority citation) is entirely absent from a dependent artefact where it is mandated by the relevant SOP	Warning	Board note template requires a delegation-of-authority citation; no such citation is present
Formatting / Reference Informational	Stylistic or reference inconsistencies that do not affect factual content (e.g. inconsistent use of abbreviations, differing currency display conventions)	Informational	NOTA writes "Rp500,000,000,000"; board note writes "IDR 500 billion"

Performance & Quality Metrics

The following metrics govern the ongoing measurement of checker accuracy and operational performance. They shall be tracked per model version and reported in the IBM delivery scorecard.

Metric	Definition	Target Threshold	Measurement Method	Notes
Check Completion Latency	Wall-clock time from check trigger to consistency report available in the user's view	<2 min (standard 4-artefact file); <4 min (extended file set)	System timestamp; P95 latency measured across all check runs in a 30-day window	Standard = 4 artefacts, each ≤30 pages. Extended = up to 8 artefacts or individual document >30 pages. Packages outside this range require a separate SLA definition.

Detection Precision (by type)	Percentage of flagged items that are genuine inconsistencies confirmed as correct upon human review (true positives ÷ all flagged items)	Overall ≥90%; Critical type ≥95%; Warning type ≥85%	Random sample of 20 completed files per month: human reviewer independently verifies each flagged item; precision computed per inconsistency type	Precision is tracked separately per type because different detection methods carry different false-positive rates. Numeric mismatch detection (deterministic) is expected to reach near-100%; semantic mismatch detection (NLP-based) will have a lower initial precision that improves over time.
Detection Recall (by type)	Percentage of actual inconsistencies present in a file that the checker successfully flags (true positives ÷ actual positives)	Overall ≥85%; Critical type ≥95%	Monthly injection test: a set of 10 controlled files with known injected discrepancies (3 Critical, 4 Warning, 3 Informational per file); recall measured against the known injection manifest	Recall on Critical types is held to a higher standard because a missed Critical inconsistency reaching a board sign-off is the highest-consequence failure mode. Missing a Critical item constitutes a test failure regardless of overall recall score.
False Positive Rate (Senior Officer View)	Percentage of flagged Critical and Warning items that the senior officer marks as "Accepted as-is" without making any document change	<15% of Critical; <25% of Warning	Ratio of "Accepted as-is" actions to total flagged items per severity level, aggregated monthly from the audit log	A rising false-positive rate increases senior officer review burden and erodes trust in the tool. If the Critical false-positive rate exceeds 20% in any rolling 30-day window, an emergency model recalibration is triggered.

Critical Item Resolution Rate Before Board Submission	Percentage of governance files that reach the board preparation stage with zero unresolved Critical items in their consistency report	100% (enforced by system gate)	Automated: system gate blocks board-stage progression if any Critical item is in "Unresolved" status; gate events logged	Any attempt to bypass the gate (e.g. via admin override) shall be logged as a compliance incident and require a dual-approval override with written justification retained in the audit trail.
Mean Time to Analyst Acknowledgement	Average elapsed time between consistency report delivery and analyst acknowledgement of the report	<4 business hours	Timestamp delta between report generation and acknowledgement action; measured per file; alerts triggered if >8 business hours without acknowledgement	Long acknowledgement delays indicate either report overload (too many false positives) or workflow bottleneck. Used as an early warning signal for precision-related issues.
Checker Coverage (Artefact Types)	Percentage of the defined artefact types in the governance file set for which the checker performs active comparison (as opposed to skipping due to format issues or missing documents)	≥95% of expected artefact slots covered per file	Per file: ratio of artefact types actually checked to artefact types expected per the relevant SOP; logged at check completion	Unchecked artefact slots (e.g. due to a missing document or unsupported format) are surfaced as Informational items in the report so the analyst is aware of the coverage gap.

Measurement Methodology

- **Injection test cadence:** Controlled injection tests (10 files with known discrepancies) shall be run monthly by the IBM QA team. Results are reviewed jointly by IBM and DAM compliance before the monthly scorecard is finalized.
- **Human review sample:** A random 10% sample of all consistency reports generated in production shall be reviewed by a designated DAM compliance reviewer each month to validate precision metrics.

Reviewer findings are logged and fed back to the IBM model team.

- **Baseline freeze:** Precision and recall baselines shall be captured from the first 30 days of pilot production before any model tuning is applied. All subsequent metrics are measured against this frozen baseline.
- **Model update gate:** Any update to the NLP-based consistency detection model must demonstrate no regression of more than 3 percentage points on precision or recall on the injection test set before being promoted to production.
- **Senior officer feedback loop:** Senior officers shall have a one-click "Incorrect flag" action on any consistency report item. These signals are aggregated weekly and reviewed by the IBM model team to identify systematic false-positive patterns for targeted model correction.

Acceptance Criteria

The following criteria must all be satisfied before FR-2 is marked as accepted at the Phase 1 go-live gate:

#	Acceptance Criterion	Test Scenario	Pass Condition	Verification Owner
A C -2 .1	Check completes within the latency SLA for a standard 4-artefact file	Trigger a consistency check on a prepared 4-artefact test file (each artefact ≤30 pages); record wall-clock time to report delivery	Report available in <2 minutes across 5 consecutive runs; no run exceeds 3 minutes	IBM QA
A C -2 .2	Numeric value mismatches (Critical type) are detected with ≥95% precision and recall	Run the checker against 10 injection-test files each containing 3 deliberately injected numeric mismatches across different artefact pairs	At least 28 of 30 injected numeric mismatches flagged (recall ≥93%); no more than 2 non-injected numeric items flagged per file (precision ≥93%)	IBM QA / DAM Compliance

A C -2 .3	Semantic inconsistencies (missing propagated updates, contradictory recommendations) are detected with ≥85% recall	Run the checker against 10 injection-test files each containing 2 deliberately injected semantic discrepancies	At least 17 of 20 injected semantic discrepancies flagged; no more than 4 non-injected semantic items flagged per file	IBM QA / DAM Senior Officer reviewer
A C -2 .4	Every flagged item includes a recommended resolution action	Inspect 3 full consistency reports from pilot test files; verify every flagged item contains a non-empty recommended resolution field	100% of flagged items carry a specific, actionable resolution recommendation; no item has a blank or generic "Review required" placeholder	DAM Compliance Officer
A C -2 .5	System gate blocks board-stage progression when unresolved Critical items exist	Attempt to advance a test file to the board preparation stage while the consistency report contains one unresolved Critical item	System denies the stage transition; displays a clear message naming the unresolved Critical item(s); no workaround is available without an admin dual-approval override	IBM QA / DAM IT

A C -2 .6	Analyst resolution actions are fully captured in the audit log	Perform "Resolved", "Accepted as-is with justification", and "Deferred" actions on flagged items in a test report; retrieve and inspect the audit log	Each action is logged with user ID, timestamp, item ID, action type, and free-text justification where provided; no records missing	IBM QA / DAM IT Audit
A C -2 .7	Automatic check triggers on senior review queue submission	Submit a test governance file to the senior review queue without manually triggering a check; verify that the check runs automatically before the file reaches the senior officer inbox	Consistency report is generated and analyst acknowledgment is required before the file appears in the senior officer's review queue; senior officer does not receive the file without a completed report	IBM QA / DAM Senior Officer
A C -2 .8	Senior officer false-positive rate is within tolerance at pilot launch	Collect resolution actions from the first 20 files processed through the checker in the pilot; compute Critical false-positive rate	Critical items marked "Accepted as-is" by the senior officer without document change: <20% of all Critical items flagged in the pilot period	IBM Delivery Lead / DAM Senior Officers

FR-3 — Version Control and Single Source of Truth

P1 — CRITICAL

User story: As a governance analyst, I want a single authoritative master document for each corporate action file with clear version history and automatic propagation of changes to dependent artefacts, so that parallel versions cannot diverge.

Functional Requirements

- The system shall maintain a master document hierarchy for each governance file, designating one document as the authoritative source for each class of data (e.g., the NOTA as master for factual submission data).
- Any edit to a master document shall trigger an automated propagation check to all dependent artefacts, surfacing sections that are now out of sync.
- The system shall maintain a full version history for every document, preserving all prior versions with metadata (author, timestamp, change summary).
- Document checkout and check-in controls shall prevent simultaneous conflicting edits by multiple users.
- The system shall support version comparison (diff view) between any two versions of a document.
- All document versions shall be immutable once submitted to the senior review queue — only a formal revision workflow can produce a new version at that stage.

FR-4 — Rule-Based Eligibility and SOP Routing Engine

P1 — CRITICAL

User story: As a governance analyst, I want the system to evaluate a corporate action request against DAM's delegation of authority and governance rules, and to tell me exactly which SOP pathway applies and which approvals are required, so that I do not route requests incorrectly.

Functional Requirements

- The system shall encode DAM's delegation of authority thresholds, governance rules, and SOP routing logic as a queryable rule engine.
- On intake of a new corporate action request, the system shall automatically evaluate the request attributes (type, value, entity, risk category) against the rule engine and output: (a) the applicable SOP, (b) the required approvals, (c) any pre-conditions that must be met, and (d) the expected processing timeline.
- Where a request clearly does not require a specific approval, the system shall flag this and suppress the unnecessary routing step.
- Where routing is genuinely ambiguous (judgment-required cases), the system shall surface the relevant rules, flag the ambiguity, and route to a designated human resolver with supporting context.
- The rule engine shall be configurable by authorized SOP owners without requiring a software release cycle.
- All routing decisions shall be logged with the rule version applied, the evaluated attributes, and the output decision.

FR-5 — Secure Enterprise AI Sandbox

P1 — CRITICAL

User story: As DAM IT security, I need all AI processing to occur within a DAM-controlled environment with no transmission of governance data to external AI providers, complete auditability of all AI interactions, and enforcement controls preventing use of public AI tools for governance work.

Functional Requirements

- All AI inference shall execute on infrastructure within DAM's controlled environment — on-premise or a designated private cloud deployment approved by DAM IT and BPI Danantara security.
- No governance document content, submission data, or user queries shall be transmitted to external AI provider endpoints or used for external model training.
- The system shall maintain a complete AI interaction log: every prompt, model version, and generated output, associated with a user identity and document file reference.
- The sandbox shall enforce role-based access controls aligned with DAM's existing organizational hierarchy.
- The system shall include a data classification layer that prevents documents classified above a defined sensitivity level from being processed by any model tier not approved for that classification.
- A real-time monitoring dashboard shall be available to IT administrators showing AI usage, data volume, model versions in service, and access anomalies.

FR-6 — Integrated Workflow and Document Orchestration

P1 — CRITICAL

User story: As a governance analyst, I want the system to guide each corporate action request through the full lifecycle — from submission intake to final communication — with automated hand-offs, checkpoints, and status visibility, so that nothing falls through the cracks.

Functional Requirements

- The system shall implement a configurable workflow engine covering the full governance lifecycle: submission intake → eligibility check → NOTA drafting → internal review → board preparation → decision → communication dispatch.
- Each stage shall have defined entry conditions, exit conditions, and assignable human owners.
- Automatic transitions shall occur for stages where all required conditions are system-verifiable (e.g., consistency check passed, all required fields populated).
- Human-required stages shall generate task notifications to the assigned owner with context, deadline, and relevant documents.
- A real-time status dashboard shall provide all stakeholders with visibility into the current stage, pending actions, elapsed time, and projected completion of each active file.
- The system shall support parallel workstreams within a file (e.g., NOTA drafting concurrent with SOP routing verification) with merge gates before downstream stages.
- Bottleneck alerting shall notify supervisors when a file has not progressed past a stage within the configured SLA window.

FR-7 — SOP Digitization and Decision Routing Engine

P2 — HIGH

User story: As a compliance officer, I want all active SOPs encoded in a machine-readable format with full decision trees and escalation paths, so that the system enforces correct process routing and my team does not need to manually interpret SOP text on each case.

Functional Requirements

- The system shall provide an SOP authoring environment where SOPs can be encoded as structured decision trees, including conditional branches, threshold rules, and escalation paths.
- Each SOP shall carry a version, effective date, owner, and approval status. Only approved SOPs shall be active in the routing engine.
- The system shall support the gradual migration of paper/document SOPs into the digital engine, with a tracking dashboard showing coverage status.
- SOP decision trees shall be visualizable by compliance officers for review and approval without requiring technical knowledge of the underlying system configuration.
- When a new submission is evaluated, the system shall record which version of which SOP was applied, creating a permanent compliance audit trail.

FR-8 — Risk-Based Automation Segmentation

P2 — HIGH

User story: As DAM senior leadership, I want the system to automatically process low-risk, straightforward corporate action requests end-to-end, while routing complex or high-risk requests to appropriate human reviewers, so that human capacity is focused where judgment is genuinely needed.

Functional Requirements

- The system shall implement a risk-scoring model that evaluates each incoming corporate action request on defined dimensions (e.g., transaction value, entity risk rating, novelty, precedent availability, regulatory exposure).
- Requests scoring below a defined threshold across all dimensions shall be eligible for fully automated processing — the system shall draft, validate, route, and produce a recommended decision without human intervention beyond final sign-off.
- Requests scoring above the threshold on any dimension shall be routed to the human-in-the-loop workflow with the risk score, flagged dimensions, and relevant precedents surfaced to the reviewer.
- Automation thresholds shall be configurable by authorized senior officers and shall require dual-approval to change.
- The system shall track the fully automated versus human-assisted processing ratio and surface this metric on the management dashboard.
- Any error, override, or exception in an automated flow shall trigger an automatic escalation to a human reviewer and shall be logged as an incident.

8 Functional Requirements — Phase 2 (Platform Expansion)

FR-9 — Enterprise Governance Platform Across BPI Ecosystem

P3 — STRATEGIC

The Phase 1 platform shall be architecturally designed for extension to the full BPI Danantara ecosystem, including DAM (Danantara Asset Management), DIM (Danantara Investment Management), and DSI (Danantara Strategic Investments).

Requirements

- The system architecture shall use a multi-tenancy model supporting entity-level data isolation with the ability to share standardized governance artefact templates, workflow definitions, and SOP libraries across entities under central BPI governance.
- A BPI-level governance dashboard shall aggregate cross-entity metrics, pending actions, and compliance status, accessible to authorized BPI central office users.
- Entity-specific customization of templates, SOPs, and routing rules shall be supported within the shared platform without impacting other entities.
- A governance artefact standards library shall be maintained centrally, with an entity adoption tracking mechanism.

FR-10 — Agentic Governance System

P3 — STRATEGIC

The long-term vision is a fully agentic governance system where AI agents autonomously handle the full lifecycle of low-risk governance matters, with human intervention triggered only by explicit rules or agent-flagged exceptions.

Requirements

- An orchestrating governance agent shall monitor all active files and autonomously trigger drafting, consistency validation, SOP routing, and status communication actions without requiring human initiation.
- A drafting agent shall produce complete governance artefact sets for eligible cases, with source traceability and confidence scores attached to each generated section.
- A validation agent shall continuously monitor artefact consistency and flag deviations in real time as documents are edited.
- A routing agent shall monitor incoming submissions and execute SOP pathway assignment automatically, escalating to human judgment only for ambiguous cases.

- Human reviewers shall interact with the agentic system via a review interface that presents only the exception cases requiring judgment, with full agent reasoning and supporting evidence surfaced.
- A full agent action log shall be maintained for every automated decision for regulatory and legal audit purposes.

9 Non-Functional Requirements

Category	Requirement	Specification
Performance	NOTA first draft generation	Complete in <5 minutes for a standard submission package (<50 pages)
Performance	Consistency check	Report delivered in <2 minutes for a 4-artefact file set
Performance	SOP routing decision	Routing output in <30 seconds from intake form submission
Security	Data residency	All governance document data processed and stored within DAM-approved infrastructure; no external transmission
Security	Model isolation	AI models deployed in an isolated compute environment; no shared inference with external tenants
Security	Access control	Role-based access control (RBAC) integrated with DAM's existing identity provider; principle of least privilege enforced
Auditability	Full action log	Every user action, system action, AI inference, and document state change logged with user identity and timestamp; tamper-evident storage
Auditability	Log retention	Audit logs retained for a minimum of 7 years in accordance with Indonesian governance and financial regulation requirements
Reliability	System availability	99.5% uptime during DAM business hours; planned maintenance windows outside business hours with 48-hour advance notice
Reliability	Data backup	Full document and audit log backup every 24 hours; point-in-time recovery within a 30-day window
Accuracy	Hallucination prevention	All AI-generated content in descriptive sections must be sourced to a retrievable passage in the input documents; unsourced generation is not permitted
Usability	Training requirement	Standard governance analyst workflow operable with <4 hours of structured training

Compliance

SOP enforcement

System shall not permit a governance file to progress past the routing stage without a recorded SOP pathway decision

10 System Architecture Principles

10.1 Core Design Principles

- **Human in the loop for judgment, system for process:** The platform shall automate all steps that are mechanical, rule-verifiable, or repetitive. All steps requiring analytical judgment, exception handling, or personal accountability shall surface to a human with full context.
- **Auditability as a first-class requirement:** Every state change, AI action, and human decision must be logged, attributable, and retrievable. Audit trail design shall be reviewed by legal and compliance before go-live.
- **Retrieval-Augmented Generation over hallucination:** All AI content generation shall use RAG against the submission package and approved precedent library. Generative content not grounded in a retrieved source passage shall be blocked.
- **Configurable rule engine, not hardcoded logic:** SOP rules, delegation thresholds, and routing logic shall be maintained in a configuration layer accessible to authorized business users, decoupled from the software release cycle.
- **Modular architecture for BPI-wide extensibility:** Core platform modules (document engine, AI sandbox, workflow engine, rule engine) shall be designed as independently deployable services that can be extended across BPI entities in Phase 2.

10.2 Integration Touchpoints

System	Integration Type	Notes
DAM document management (OneDrive migration)	Data migration + ongoing sync or replacement	Document governance policy must be established as part of migration
DAM identity provider / Active Directory	RBAC / SSO	Required for access control and audit log attribution
SOE subsidiary submission portals	Submission intake API or secure upload portal	Structured submission format to be agreed with subsidiaries
Board management system (if present)	Board note export / board pack integration	To be determined in technical discovery
IBM enterprise AI infrastructure	Private model deployment (IBM watsonx or equivalent)	Must meet DAM data residency and isolation requirements

11 Data & Security Governance

11.1 Data Classification

All governance documents processed by the platform shall be classified at intake. The data classification schema shall be aligned with BPI Danantara's institutional policy and applied to determine permissible AI model tiers, storage locations, and access roles.

11.2 AI Governance Controls

- No AI-generated content shall be incorporated into a governance artefact without a traceable link to a source passage and an explicit human acceptance action.
- Model versions used for all AI inference shall be recorded in the audit log; no model version shall be changed in a production environment without a documented approval.
- AI confidence scores shall be surfaced alongside generated content to assist analysts in prioritizing manual review effort.
- A red-team / adversarial testing cycle shall be conducted before go-live and at each major model update to evaluate hallucination rates and boundary cases.

11.3 Transition from Public AI Tool Use

The deployment plan shall include a formal policy transition: the use of public AI tools (e.g., ChatGPT, Copilot with external connectivity) for governance document work shall be formally prohibited once the secure AI sandbox is operational. A technical control (network-level block on submission content to external endpoints) shall be implemented in coordination with BPI Danantara IT security.

12 Risk Register

#	Risk	Severity	Likelihood	Mitigation
R 1	AI hallucination in governance artefacts creates a flawed decision	Critical	Medium	RAG-only generation with mandatory source traceability; human acceptance gate on every AI paragraph; red-team testing cycle
R 2	Sensitive governance data leaks to external AI provider	Critical	Low (post-sandbox)	Fully isolated private AI deployment; network-level controls; no external API calls from model inference layer
R 3	SOP rule engine misconfiguration causes incorrect routing	High	Medium	Dual-approval on rule changes; simulation test environment for new rules before production activation; audit log of all routing decisions
R 4	User resistance to AI-assisted workflow reduces adoption	Medium	Medium	Change management program led by senior DAM sponsors; pilot with willing analysts; productivity metrics shared early
R 5	Incomplete SOP digitization leaves gaps in the routing engine	Medium	High	Track SOP coverage dashboard from day one; treat ambiguous routing as a human-resolver case until the relevant SOP is encoded; SOP digitization as explicit Phase 1 workstream
R 6	Version control discipline lapses — teams bypass system and use file shares	Medium	Medium	System becomes the only authorized intake channel; disable OneDrive-based collaboration for governed document types; workflow gate requires system-generated document IDs
R 7	Personal liability concerns prevent officer sign-off on AI-assisted artefacts	High	Medium	Legal opinion on AI-assisted document governance; clear attribution of all AI vs human content; policy statement from DAM leadership on institutional accountability framework

13 Out of Scope

- Origination and preparation of SOE subsidiary submission documents — the platform receives submissions as inputs, it does not assist subsidiaries in preparing them
- Legal drafting of formal regulatory filings — the platform supports internal governance artefacts, not external regulatory submissions
- Financial modeling, valuation, or quantitative analysis of proposed corporate actions — these analytical functions remain with specialist teams
- Replacement of any human judgment section in governance documents — the platform explicitly preserves human authorship of all analytical, risk, and recommendation content
- Public-facing communication channels or investor relations materials
- Phase 2 entity onboarding (DIM, DSI) — to be scoped in a separate Phase 2 PRD following Phase 1 delivery

14 Open Questions & Assumptions

14.1 Open Questions

#	Question	Owner	Target Resolution
Q 1	What is the preferred infrastructure deployment model — on-premise, private cloud (AWS/Azure dedicated), or IBM managed private cloud?	BPI Danantara IT / IBM	Technical discovery phase
Q 2	Which IBM AI model(s) are approved for the secure sandbox deployment (watsonx Granite, fine-tuned variant, or other)?	IBM Delivery Lead	Architecture design phase
Q 3	What is the complete inventory of active SOPs? Which SOPs are highest-priority for Phase 1 digitization?	DAM Legal / Compliance	Pre-Phase 1 kickoff
Q 4	Is there a legal opinion supporting the use of AI-generated content in formally signed governance artefacts?	DAM Legal	Before Phase 1 pilot go-live
Q 5	What is the format and submission protocol for SOE subsidiary corporate action requests? Is there a standard submission template?	DAM / SOE Relations	Pre-Phase 1 kickoff
Q 6	Who are the pilot users for the Phase 1 quick-win deployment, and what governance file types will be included in the pilot?	DAM Senior Leadership	Pre-Phase 1 kickoff

14.2 Key Assumptions

- DAM will make available a corpus of historical approved NOTA documents and related governance artefacts for RAG model grounding and fine-tuning, subject to appropriate data governance controls.
- BPI Danantara IT will provide a dedicated infrastructure environment meeting the data residency and isolation requirements defined in FR-5 within the Phase 1 timeline.
- DAM senior leadership will formally sponsor the transition away from public AI tool use and communicate this policy change to all governance staff upon sandbox go-live.
- The 30% productivity improvement target applies to overall governance document preparation time and is achievable through the Phase 1 feature set without requiring Phase 2 agentic capabilities.
- SOP ownership and authority to encode SOP rules in the digital engine is formally delegated to an identified DAM compliance officer role before Phase 1 rule-engine configuration begins.

15 Appendix — Feature Inventory & Prioritization

ID	Feature	Phase	Priority	Primary Pain Point Addressed
FR-1	AI-Assisted Drafting Engine	Phase 1	P1	Pain D (repetitive manual drafting); Pain E (uncontrolled AI)
FR-2	Cross-Document Consistency Checker	Phase 1	P1	Pain A (multiple versions of truth); Pain B (over-documentation)
FR-3	Version Control + Single Source of Truth	Phase 1	P1	Pain A (inconsistency); Pain B (redundant duplication)
FR-4	Rule-Based Eligibility and SOP Routing Engine	Phase 1	P1	Pain C (fragmented workflow); Pain F (personal liability)
FR-5	Secure Enterprise AI Sandbox	Phase 1	P1	Pain E (hallucination, confidentiality, auditability)
FR-6	Integrated Workflow and Document Orchestration	Phase 1	P1	Pain C (fragmented workflow); Pain A (version control)
FR-7	SOP Digitization and Decision Routing Engine	Phase 1	P2	Pain C (unclear SOP routing); Pain F (codify rules into system)
FR-8	Risk-Based Automation Segmentation	Phase 1	P2	Pain D (repetitive work); Pain F (remove human from low-risk processes)
FR-9	Enterprise Governance Platform — BPI-wide	Phase 2	P3	Strategic: BPI ecosystem standardization
FR-10	Agentic Governance System	Phase 2	P3	Strategic: full workflow automation at scale